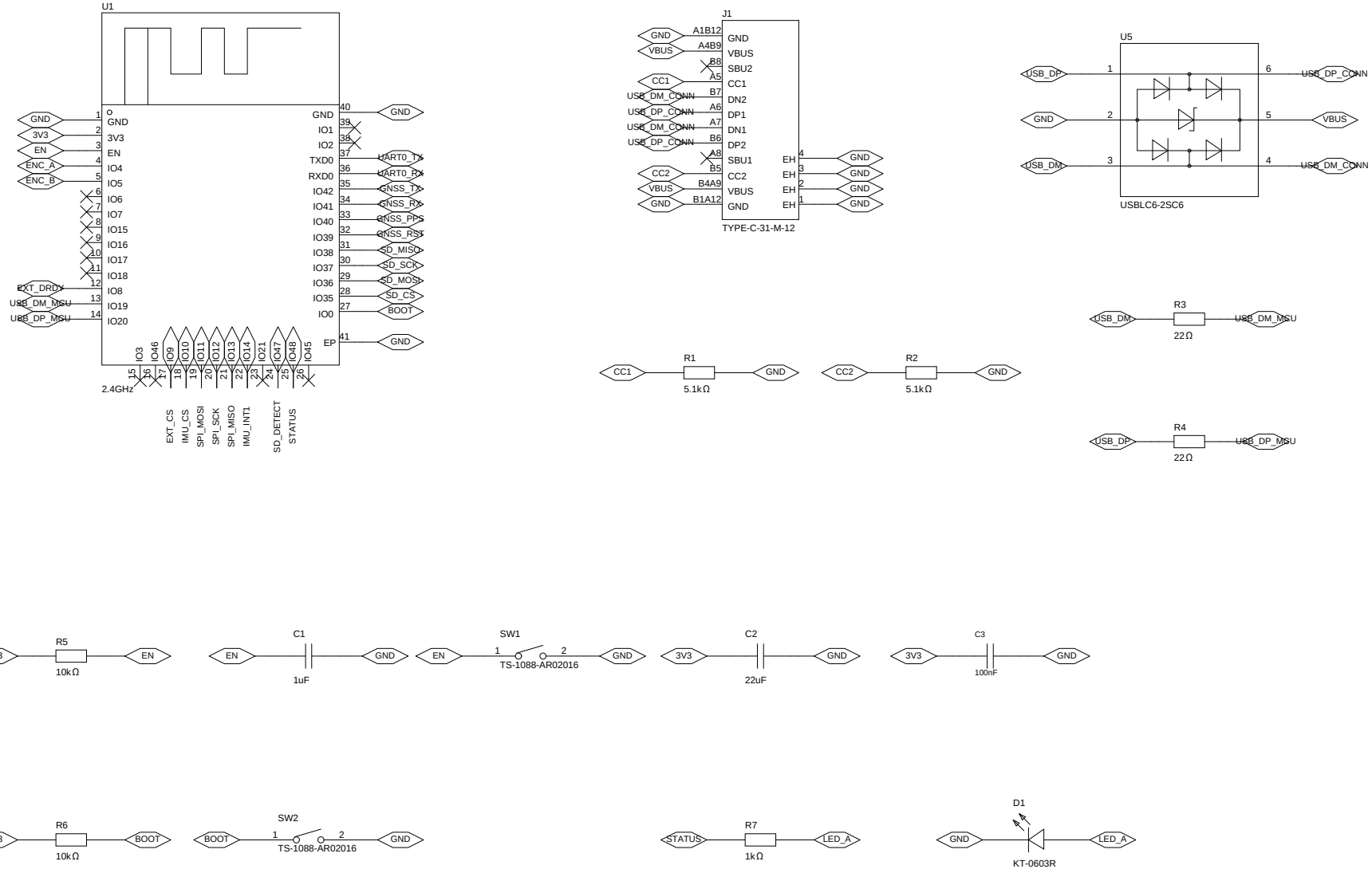


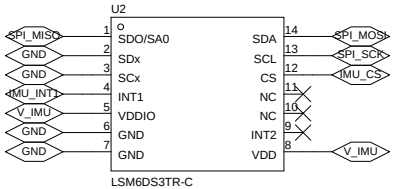
01 MCU / USB / BOOT - R2

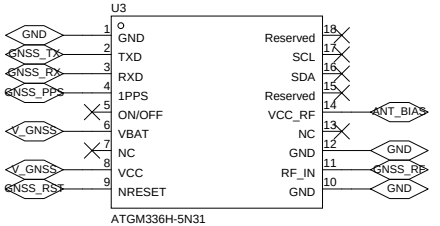
ESP32-S3-WROOM-1-N8 (no PSRAM); USB native FS

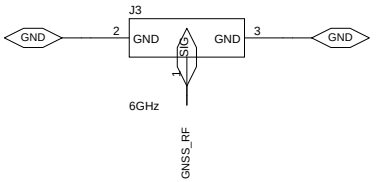
BOOT: hold SW2, tap SW1 RESET, release SW2

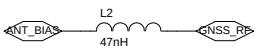
PCB bottom TP1=TX0 TP2=RX0 TP3=EN TP4=BOOT

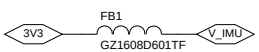
USBLC6: pins 1-6 and 3-4 connect internally; connector nets are split

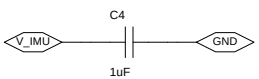


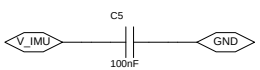




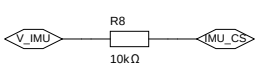


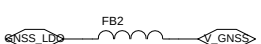


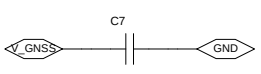












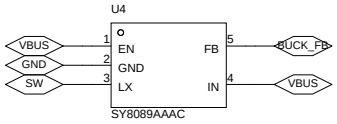


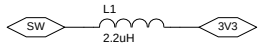
02 IMU / GNSS - R2

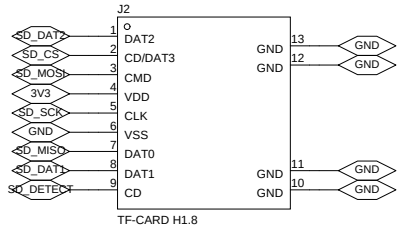
LSM6DS3TR-C SPI + INT1 / FIFO; ATGM336H-5N31 UART + PPS

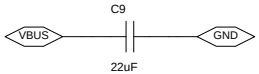
J3: 3.3 V active antenna, prefer 20 mA or less; 50 mA is short-circuit limit

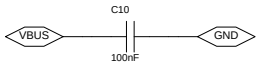
V_GNSS: dedicated 3.3 V LDO through FB2; VBAT shares supply

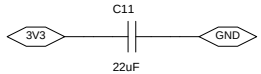


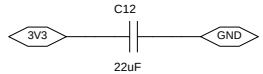




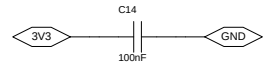


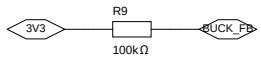


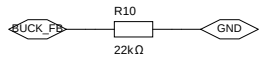


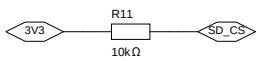


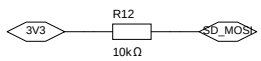


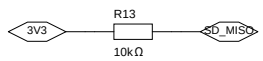


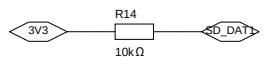


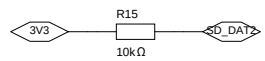


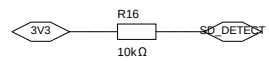












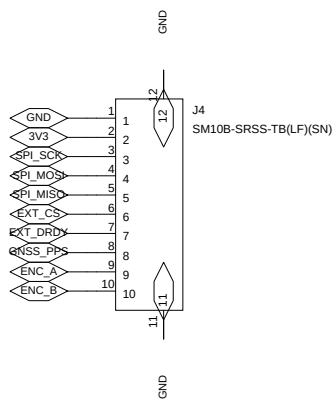
03 USB POWER / microSD - R2

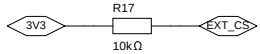
USB 5 V; buck output 3.327 V nominal; C9 now 22 uF

39 x 43 mm; four copper layers; 52 components / 25 part types, all top

5 assembled boards; TF card and antenna external

R2

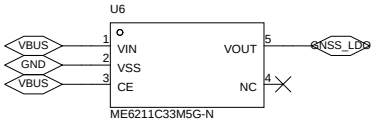


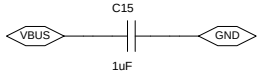


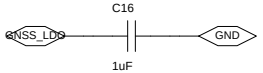
04 EXPANSION / 3.3 V LOGIC - R2

J4: 1 GND 2 3V3 3 SCK 4 MOSI 5 MISO

6 EXT_CS 7 EXT_DRDY 8 PPS 9 ENC_A 10 ENC_B











05 GNSS POWER / PROTECTION - R2

U6: ME6211C33M5G-N; C15 input and C16 output before FB2

D2: PESD5V0F1BL (0.4 pF typ), antenna-port ESD shunt

C17: 100 nF local USBLC6 VBUS bypass; all labels connect across sheets